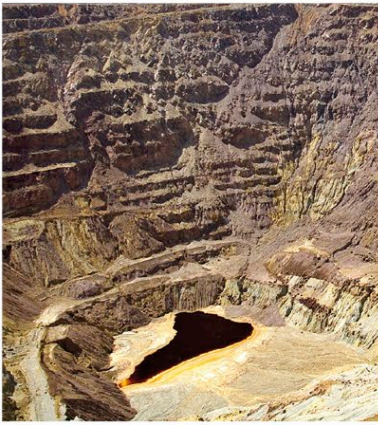




^ SCARRED LANDSCAPE

The growth of industry has required the exploitation of raw materials. Their extraction, such as at this copper mine, has changed global landscapes for ever.



^ ATMOSPHERIC POLLUTION

Slash-and-burn forest clearing for agriculture not only releases atmospheric pollutants but also diminishes the capture of carbon dioxide by the plant life.

✓ GREENHOUSE EFFECT

An excess of greenhouse gases in the atmosphere creates a shield which prevents some of the Sun's energy from radiating back into space

HUMAN IMPACT

The rapid growth of the human population has had an enormous impact upon Earth's natural environments, affecting the climate and countless species of plant and animal life. Some of these changes are irreversible.

ENVIRONMENTAL CHANGE

Earth has a long history of climate change with dominant warm "greenhouse" climates, widespread forestation, and no polar ice punctuated by glacial "ice ages". Global warming is known to be linked to high levels of greenhouse gases in the atmosphere, such as carbon dioxide and methane, which trap incoming solar energy and raise the temperature of the oceans, land, and air. In the past, natural increases in atmospheric carbon dioxide have been balanced by the development of forests on land and lime-rich sediments in the seas, which eventually became coal and limestone and stored excess carbon dioxide effectively. Since the industrial revolution of the 19th century, human activity has released huge amounts of carbon dioxide and other greenhouse gases back into the atmosphere through the mining and burning of fossil fuels, the clearing of forests, and the rearing of cattle.

THE OCEANS

The health of Earth's oceans is vital to all life. Marine life depends upon the circulation of oceanic waters that contain enough oxygen and nutrients to support the food chain from plankton and shellfish to all the other animals that depend upon them for food. Fossil records show that when the oceanic environment deteriorated in the past, it led to extinctions of life. Today, over-fishing and pollution, especially with plastics, by humans are affecting the marine environment.

THE ATMOSPHERE

For thousands of years human activity has affected the atmosphere. Initially, it was restricted to the release of pollutants from domestic fires and forest clearance. In Roman times metal production released the first industrial pollutants into the atmosphere, leaving traces that can be seen in polar ice cores. Over the last 200 years, pollution by gases and particulates has risen steeply. This has produced acid rain and smog, and greenhouse gases that are linked to global warming and depletion of the ozone layer, which helps screen out harmful ultraviolet radiation.

THE LAND

Since settlements and agriculture became widespread 8,000 years ago, humans have had a growing impact upon Earth's landscapes. Worldwide population growth, land settlement, and demand for food has left few untouched landscapes. Greater awareness of the impact of human activity on the environment is now leading to efforts to conserve natural habitats.



✓ POLAR ICE SHELF COLLAPSE

Rising temperatures are leading to the collapse of the polar ice shelves. The release of such large volumes of water is raising sea levels, which in turn threatens coastlines.

